VAN ALEN INSTITUTE

PROOF-OF-CONCEPT

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EXECUTIVE SUMMARY

The Van Alen Institute is a nonprofit organization focused on inclusive urban design. The organization is launching two community-led public art installations in Spring 2025. In order to measure the impact of these installations, they need a structured data collection system. In their current system, their engagement data is gathered inconsistently, which limits the ability to derive meaningful insights.

PROJECT INTRODUCTION

Problem Statement

The Van Alen Institute (VAI) currently uses manual or third-party methods to collect participant feedback on installations. These methods may require additional resources, be time-consuming, and lack efficient real-time reporting.

Solution Statement

The community engagement system project aims to develop a web-based application that enables real time data collection, analysis, and visualization. The system will allow stakeholders to measure their impact, generate reports, and support future funding efforts. This project entails secure storage and user-friendly design that will ensure accessibility for both researchers and community members. By implementing this new system, the Van Alen Institute will gain a reliable tool to evaluate public art initiatives and continue their mission of equitable urban development.

Given all tools used in the project are open-source and free, makes this project scalable, and allows for flexible growth. The system has no direct financial burden and remains feasible within our current skill and time constraints. Additionally, these open-source technologies ensure the application is compatible with most current and legacy operating systems, enhancing adaptability across different environments. This solution minimizes reliance on proprietary tools, improves data accuracy, and equips stakeholders with real time insights to support informed decision-making and strategic planning.

Alternative Solutions

The team also carefully considered technical skillsets, time, and budget constraints. As we weighed the pros and cons of several alternatives, the following ideas were reviewed:

- Mobile Application with Location Tracking: This project would implement a mobile application published in Google and Apple's App Store. The mobile application would identify the Installation based on a participant's location and track engagement with the installation based on geolocation, time tracking and nearby NFC or Bluetooth devices. The application would automatically record and store this data on a database and process participants engagement at the installation. This option was deemed the least feasible due to implementation challenges, and privacy concerns. It would require additional hardware at installation sites, developer licensing, and approval from app stores like Apple and Google. Furthermore, adoption of mobile apps is often low without added engagement features, and location tracking raises legal and ethical concerns around user consent and surveillance.

- Data Capture System for Participants: This solution would aim to implement an engagement platform installed at the art installations for community and nearby participants to provide their own feedback and engagement with the location. The project would require an independent hardware smart mobile

device installed along with the art installation where individual would input their feedback. While this type of system could be valuable, it would require staff or volunteers to monitor and secure the device, necessitating training and potentially contradicting VAI's staffing limitations.

OUR RESEARCH

Group 1 created a survey for the Van Alen Institute, considering its interests and the proposed system. The questions are based on past surveys by the Institute, reflecting trends and data they have researched, with adjustments for new installations at Washington Heights and Gowanus.
The survey length ensures essential data while being reasonable for participants. According to SurveyMonkey, longer surveys lead to "satisficing," where participants give satisfactory but not optimal answers, compromising data quality.

The survey is organized logically by topic to keep it engaging. Ideally, surveys should take no more than 10 minutes to maintain participant interest and response rates.

Open-ended responses were eliminated as "forced-choice questions tend to yield more accurate responses" (Pew Research Center). The option "Other" remains for flexibility without allowing open-ended responses. Questions focusing on participant experiences were added, breaking down sentiments into individual ratings for structured feedback.

Numerical responses replaced agree-disagree formats due to acquiescence bias, where participants tend to agree with statements. Numerical scales provide clearer interpretations of participant reactions.

For future surveys, Group 1 suggests multi-language options to increase response rates and improve data quality, especially reflecting the neighborhoods of the installations.

DRAFTED SURVEY QUESTIONS

1. Time Spent at the Site

Q1: Before this installation, how often did you visit this site?
- ☐ Daily
- ☐ A few times a week
- ☐ Once a week
- ☐ A few times a month
- ☐ Rarely/Never

Q2: Since the installation, how often do you visit this site?
- ☐ Daily
- ☐ A few times a week
- ☐ Once a week
- ☐ A few times a month
- ☐ Rarely/Never

Q3: On average, how much time do you spend at this site per visit?
- ☐ Less than 5 minutes
- ☐ 5-15 minutes
- ☐ 15-30 minutes
- ☐ 30 minutes to 1 hour
- ☐ More than 1 hour

2. Sentiment About the Site

Q4: How welcome does this site make you feel? (In a scale from 1 to 5, 5 being very welcome) ☐ 5 ☐ 4 ☐ 3 ☐ 2 ☐ 1

Q5: How comfortable does this site make you feel? (In a scale from 1 to 5, 5 being very comfortable)

☐ 5 ☐ 4 ☐ 3 ☐ 2 ☐ 1

Q6: How safe does this site make you feel? (In a scale from 1 to 5, 5 being very safe) ☐ 5 ☐ 4 ☐ 3 ☐ 2 ☐ 1

Q7: On a scale from 1 to 5, how positive is your overall experience at this site? (5 being very Positive.)

☐ 5 ☐ 4 ☐ 3 ☐ 2 ☐ 1

3. Experience at the Site

Q8: What activity best describes your time spent at this site?

☐ Read materials on-site
☐ Played or engaged with the installation
☐ Took a phone call
☐ Ate or drank
☐ Socialized with others
☐ Passed through without stopping
☐ Other

4. Awareness of the Installation

Q9: How did you first hear about this project?

☐ Social media
☐ Word of mouth
☐ Walking by
☐ Community event
☐ Local news or website

Q10: Has this installation made you more interested in exploring the surrounding neighborhood?

☐ Yes
☐ No
☐ Not sure

5. About you

Q11: What is your age group?

☐ Under 18
☐ 18-24
☐ 25-34
☐ 35-44
☐ 45-54+

Q12: What is your gender?

☐ Male
☐ Female
☐ Non-binary
☐ Prefer not to say
☐ Other

Q13: What is your race/ethnicity? (Provide multiple-choice with an option to select multiple.)

☐ Black or African American
☐ Hispanic or Latino/a/x
☐ White
☐ Asian
☐ Native American or Alaska Native
☐ Native Hawaiian or Other Pacific Islander
☐ Middle Eastern or North African
☐ Multiracial
☐ Prefer not to say
☐ Other:

Q14: Are you a resident or member of this community?

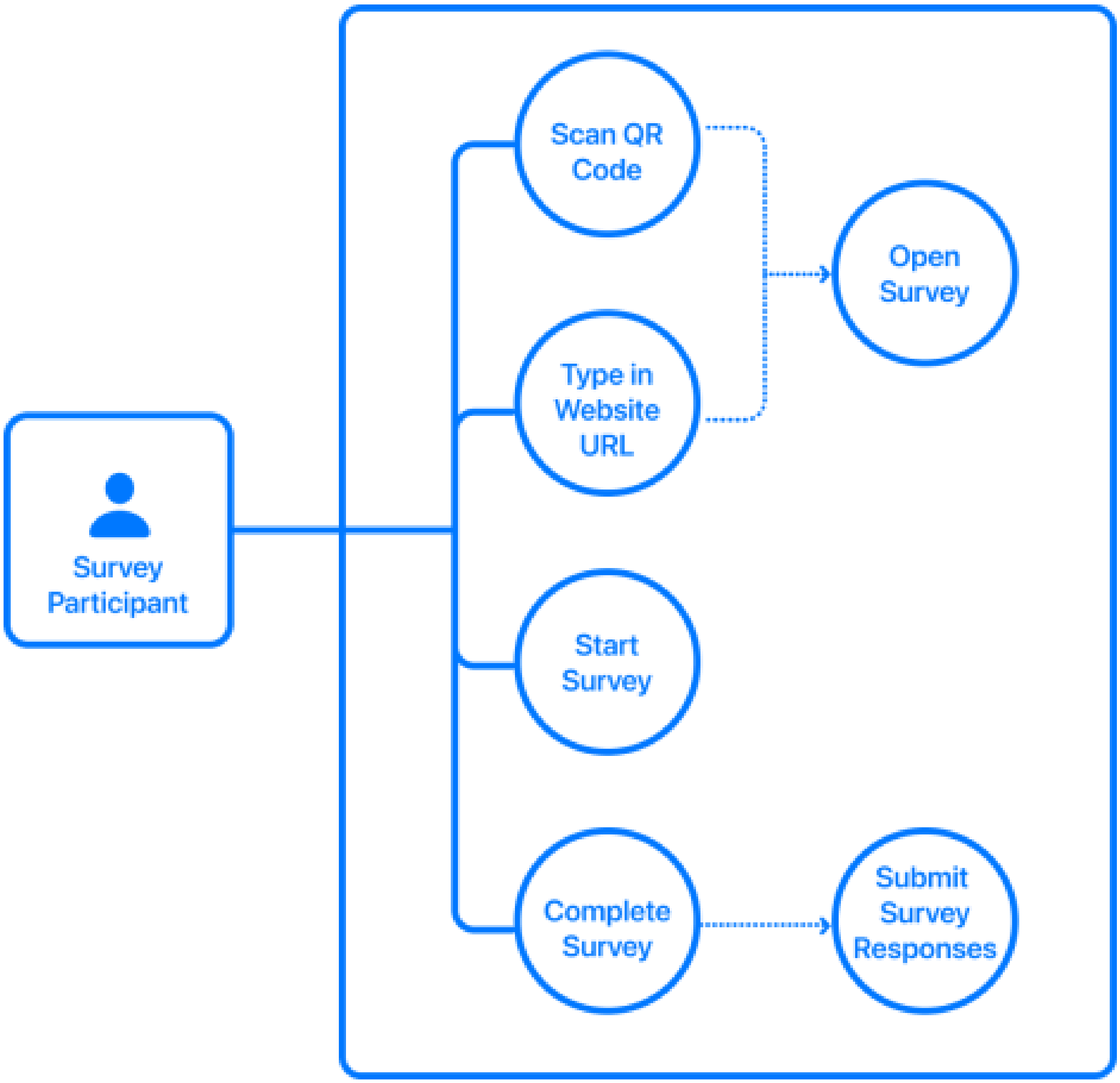
☐ Yes, I reside in this area and/or am part of this community
☐ No, I live outside this area

PRODUCT BACKLOG

ID	ITEM	USER STORY	ESTIMATE (DAYS)	PRIORITY	SPRINT	STATUS
1	Develop Web-based Interface	As a community participant, I want a simple and easy-to-use interface that I can use to provide my feedback on the art installations.	6	High	1	Started
2	Create Web - Application Wireframes	As a Project Sponsor, I want to visualize the project and provide feedback on the design of the application.	5	High	1	Completed
3	Draft Survey Questions	As a Project Sponsor, I want a survey that is engaging, easy to adopt, effective and data driven.	3	High	1	Completed
4	Design User Interaction Diagrams	As a Project Sponsor, I want to understand and visualize how participants will engage and provide data.	3	High	1	Completed
5	Analyze Sample Data	As a VAI Administrator, I want to understand the data collected by a third party to understand the effectiveness of the method and identify key features.	10	High	1	Completed

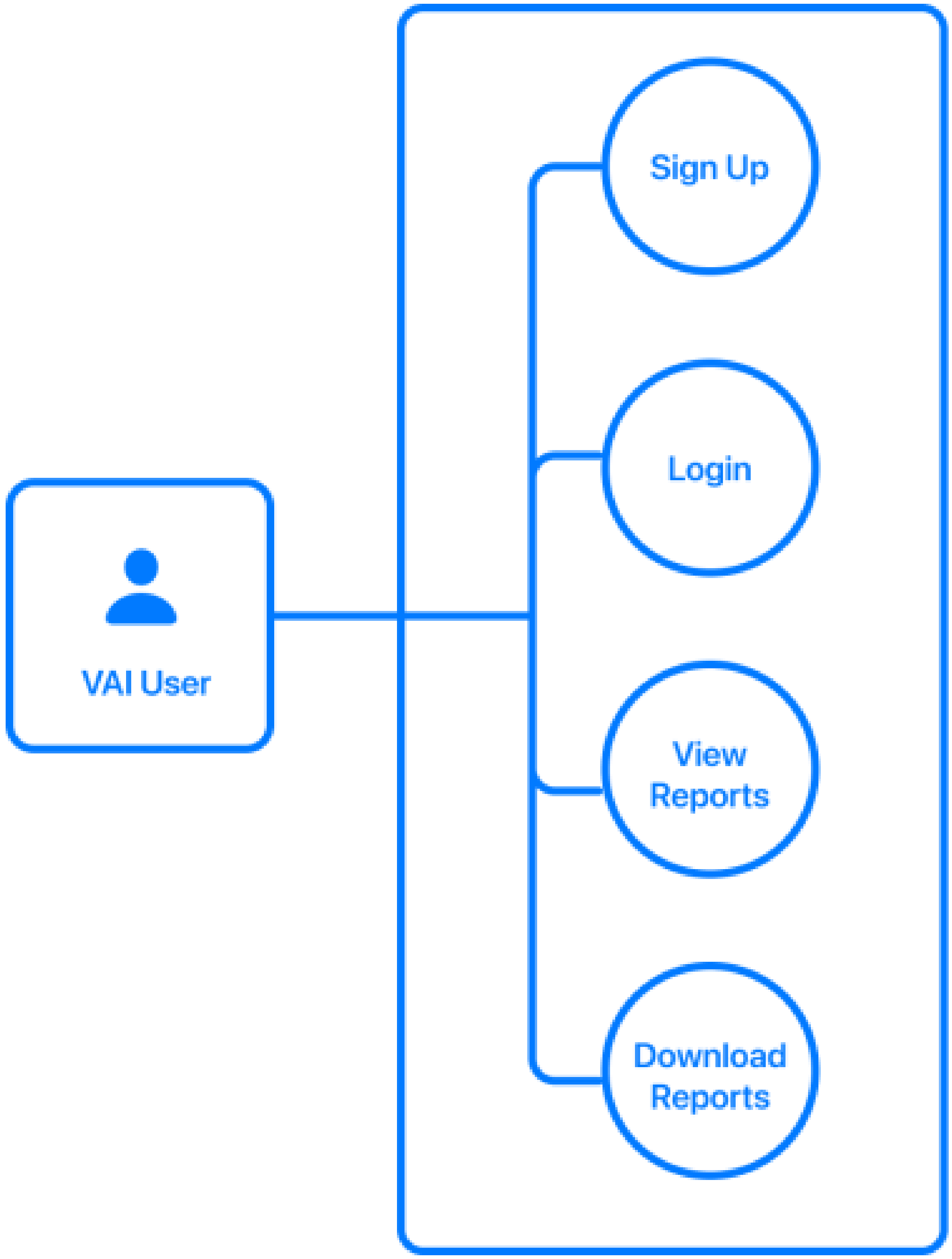
6	Develop an Online Data Visualization Interface	As a Van Alen admin, I want to be able to have visual representations of an installation's performance remotely.	10	Medium	3	Completed
7	Database Implementation	As a Van Alen Admin, I want the survey responses to be stored and processed in a remotely accessible and secure location.	3	High	1	Completed
8	Security Implementation	As a Van Alen admin, I want a secure login authentication method to ensure my account is better protected from unauthorized access. The security should extend to our database systems to ensure that the data collected from events are secured and can only be seen by Van Alen admins.	10	High	2	Completed
9	Process Data for Statistics and Analytics	As a Van Alen Financial Partner, I want to understand and visualize the impact of installation for future business decisions	18	High	3	Completed

USE CASE DIAGRAM OF A PARTICIPANT ENGAGING WITH THE SURVEY SYSTEM



* Once a community member submits the completed survey, the survey data will be automatically saved into the system database.

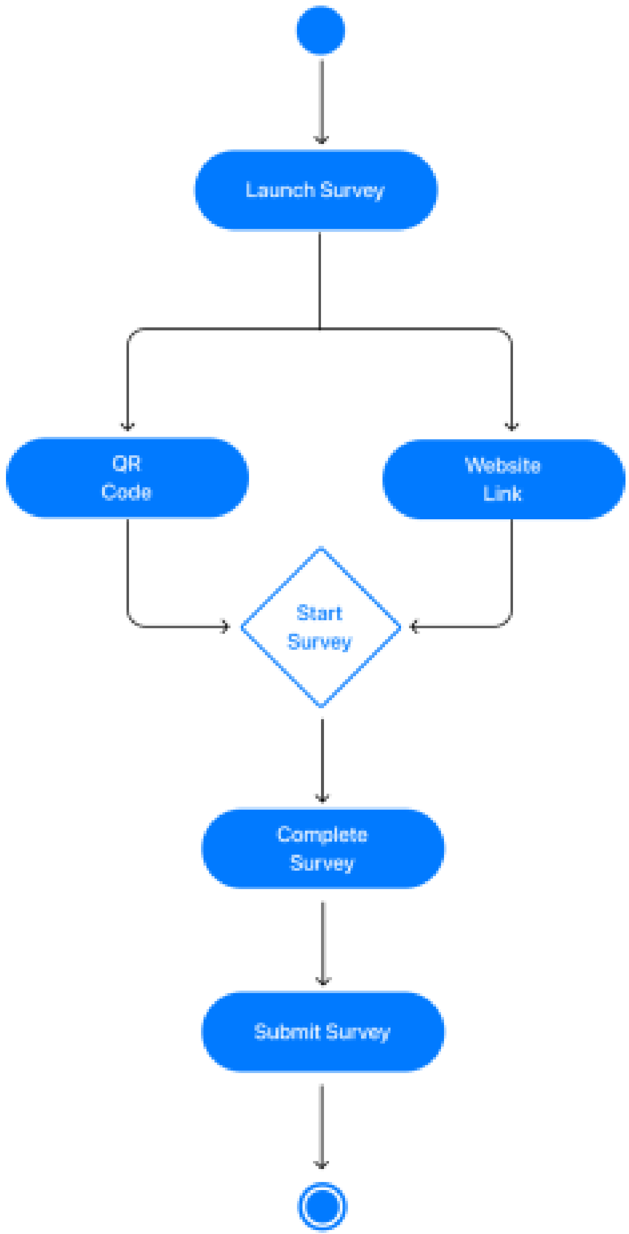
USE CASE DIAGRAM OF VAI STAFF ENGAGING WITH THE SURVEY SYSTEM



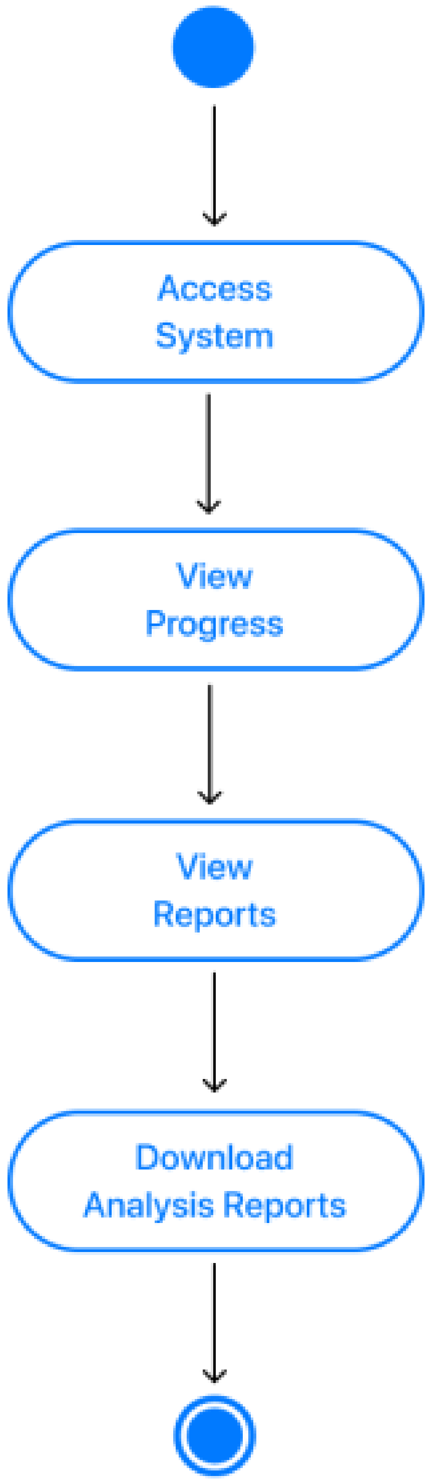
* *

Assumption: Designated VAI staff will have an account and be signed in to access the Survey System.

ACTIVITY DIAGRAM OF THE PARTICIPANT ENGAGING WITH THE SURVEY SYSTEM



ACTIVITY DIAGRAM OF VAI STAFF ENGAGING WITH THE SURVEY SYSTEM



PROPOSED WIREFRAMES FOR WEB PAGES VAI STAFF VIEW

Welcome Screen

Includes three options:

- Login
- Sign Up
- Get Started

Sign Up

Designated VAI Member will be able to create an account.

Login

Once an authorized VAI Member has an existing account, the staff member will login with email and password credentials

Dashboard

Displays live survey results with different options:

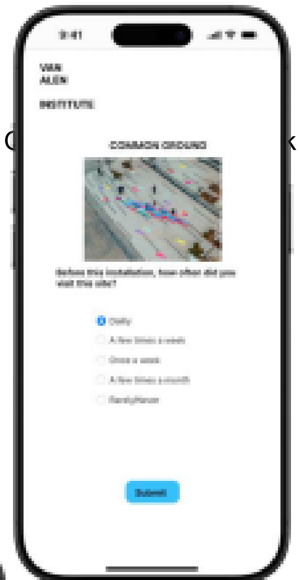
- Pie
- Donut
- Bar

** Dashboard offers a “Download” option to save reports and graphs.

PROPOSED WIREFRAMES FOR MOBILE-FRIENDLY

WEB PAGES PARTICIPANT VIEW Welcome Screen Art

Installation Screen



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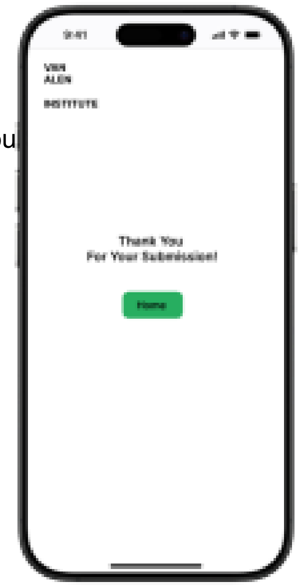
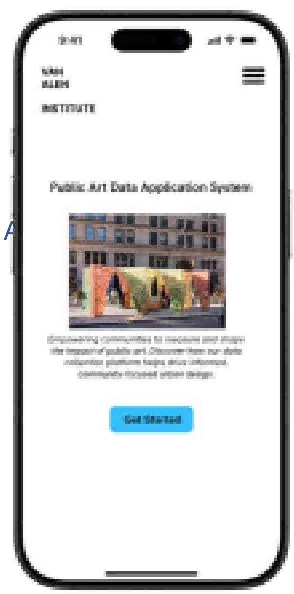


Welcome Screen



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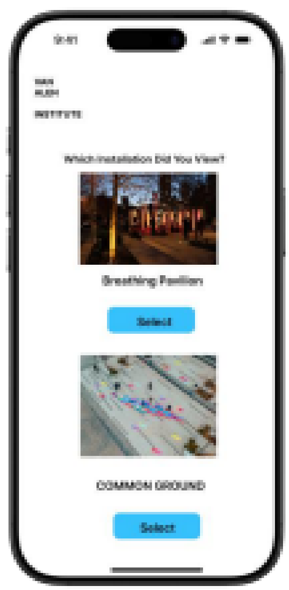


Participants can begin a survey once they press on

• Get Started

Questions Screen

Participants can select their answers.
A survey is recorded once the individual presses

• Submit

Participants start survey by selecting the Installation they're located.

The participant will be directed to a Thank You page to exit and return to the Home Page

Thank You Screen

CONCLUSION

Our proposal presents a prototype for a community data collection system designed to automate data reporting, gather community sentiment regarding art installations, and measure the impact of the Van Alen Institute's initiatives. This system aims to support data-driven decision-making and help raise funding for future projects. It features a web-based, mobile-friendly application with a simple, intuitive user interface and options for downloading automated reports. The system is built with future adoption and scalability in mind, utilizing an Object-Oriented approach to accommodate the Van Alen Institute's continued growth.

The prototype delivers a data collection system in the form of an online survey, which community participants can easily access once an installation has been erected. This can be facilitated by promoting the installation at its location and including a QR code that directs participants to the survey, where they can provide their sincere feedback regarding the installation. As detailed in our research, the survey questions were derived from past projects but have been adjusted to create a more concise and effective communication style, maximizing response rates while being mindful of participants' time constraints.

The VAI private integration ensures a secure signup and login system, accessible only to designated staff members. Once administrative access has been granted, each authorized user will have a dedicated login to view live results from an installation's performance, derive meaningful insights from the data, and download automated reports — available in a ready-to-print format and as a CSV file summarizing the graphical data.

Throughout the development process, we prioritized feedback and aimed to minimize unnecessary communications by delivering a comprehensive, reliable solution with minimal need for iterations. Our goal was to create a system that the organization can implement with confidence and derive meaningful value from. We hope this project serves as a strong alternative to the Institute's current data collection methods and addresses the challenges faced in gathering accurate and timely community feedback.

Attached to this proposal is a user guide detailing how to install and deploy the system for public use, along with instructions outlining all system features and offering additional suggestions for future enhancements.

QUESTIONS

The community engagement application began as a collective project built from the ground up based on the ideology of feedback. Our team has prepared a series of questions that we believe will be valuable to the continuity of the project and ensure we continue to address the evolving needs of the organization. VAI's feedback is imperative to improving the system's development, security, and user experience.

1. Dashboard Access and Security Restrictions

The web application has implemented security measures to ensure that only dedicated VAI staff members can access the Dashboard System. We included password requirements that must be met during account creation. In earlier versions of the system, we restricted sign-up capabilities to emails ending in @VAI.org as an example of how access restrictions could be managed. However, we believe that the leadership at VAI should ultimately determine the preferred access policy.

Question: Who should have access to the Dashboard? How would you like us to configure security restrictions regarding dashboard access?

2. Survey Structure and User Flow Preference

Currently, the application follows a linear structure where participants access the website, select the art installation they are visiting, and complete the survey.

In earlier drafts, we considered a hierarchical structure where each installation would have its own dedicated URL, automatically directing participants to the corresponding survey (removing the need for manual selection).

Question: Do you have a preference between the current linear structure and the alternative hierarchical structure? If so, which structure do you prefer, and why?

3. User Interface Design Considerations

An earlier version of the project included an alternative user interface design (attached on the next page). Due to time constraints, this design was not implemented.

Question: Should VAI decide to update the interface in the future, would you prefer a design that utilizes more whitespace for a cleaner, minimalist look, or would you prefer to fully utilize available space without intentional blank areas?

4. Survey Content Feedback

Our goal was to create a data-driven solution that efficiently supports VAI's data collection needs. We performed extensive research on language efficacy and survey structure, and based on that, we curated and adjusted questions drawn from previous projects.

Questions:
• Are there any additional questions you would like to see included in the survey?
• Do you believe the current set of survey questions captures the information VAI needs? 5.

Language Accessibility for Future Enhancements

Although we were unable to implement multilingual support due to time constraints, we believe this would be a meaningful enhancement for future versions of the application.

Question: Are there any specific languages VAI would like to prioritize to make the survey more accessible to non-English speakers? If so, which languages?

SYSTEM INSTALLATION

This application allows community members who have experienced the art installations to navigate to the site and complete a survey. Each survey submission is automatically stored in the Firestore database. When a VAI Admin logs into the platform, they can view all community responses retrieved directly from the database. Additionally, VAI Admins have the ability to generate reports that summarize the collected data, providing both visual summaries and downloadable files.

Video Demo

VAI Admin Perspective
Community Member Perspective

VAI-DataSystem | Local Setup Instructions

Prerequisites

Install the following tools:

- Node.js and npm
    - Install via homebrew (recommended) - https://formulae.brew.sh/formula/node
    - Additional Guides:
        - Download Node.js and npm
        - npm installation guide
- Python 3 (latest version)
    - Download Python
- Git
    - Download Git

You will also need:

- A Google account (to set up the Firestore database)
- A GitHub account (to access the repository)

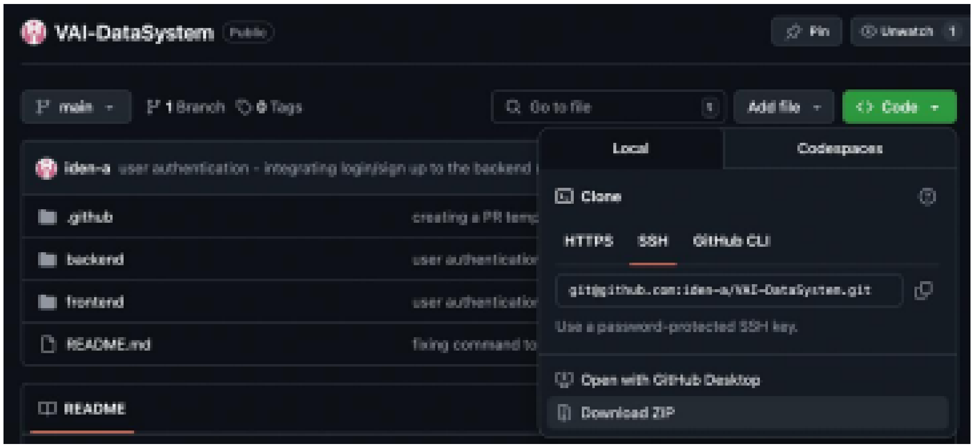
Setting up the Project Locally

Sign-in to GitHub or click on the link below and login to GitHub

Repository: https://github.com/iden-a/VAI-DataSystem
Click on the 'Fork' button in GitHub
Navigate to the repository you have forked, it should be under your 'repositories' section

1. Open your Terminal (on Mac) / Command Line (on Windows) and type in:

```
git clone git@github.com:your-username/VAI-DataSystem.git
```



cd VAI-DataSystem

Frontend Setup

1. Navigate to the frontend folder:
   cd frontend
2. Install frontend dependencies:
   npm install
3. Create a .env file:
   touch .env
4. Navigate to the .gitignore file and make sure your .gitignore includes:
   **/.env

If you accidentally push your .env to GitHub, remove it from tracking:

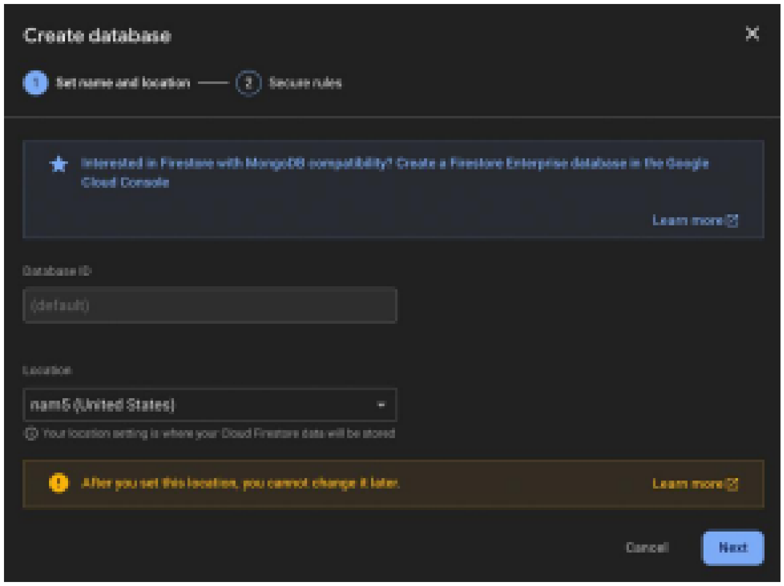
```
git rm --cached frontend/.env
git commit -m "remove .env from tracking"
git push
```

5. Run the frontend server:
   npm run dev

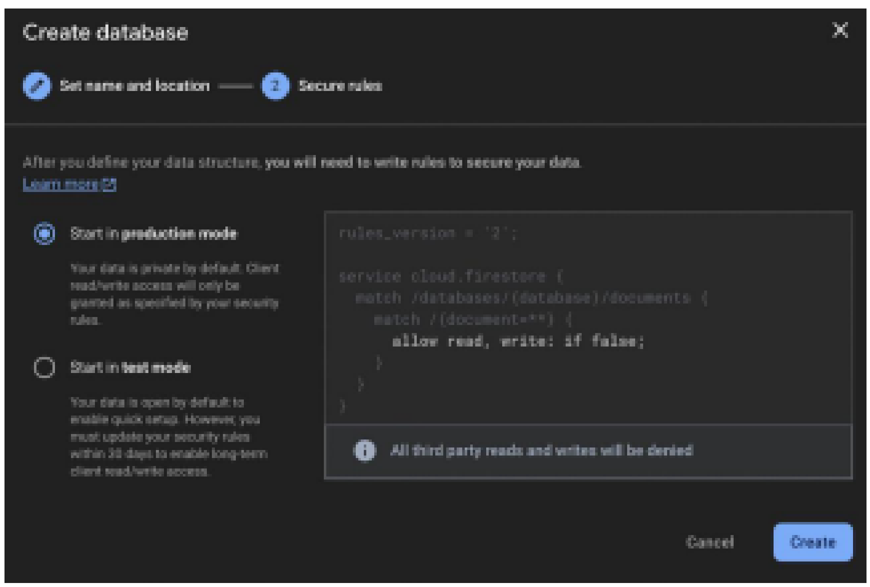
Database Setup (Firebase)

Video Guide (Recommended up to 3:35): [Watch here]

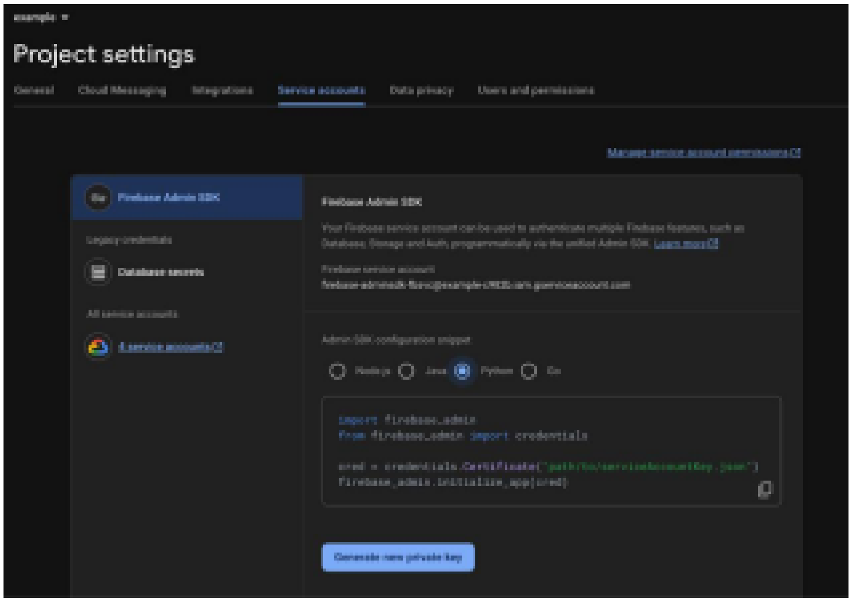
1. Create a new Firebase project: [Firebase Console]
   - Project name: VAI-DataSystem
   - Use default settings
2. Set up Firestore:
   - Click "Build" > "Firestore Database".
   - Click "Next" with default settings.



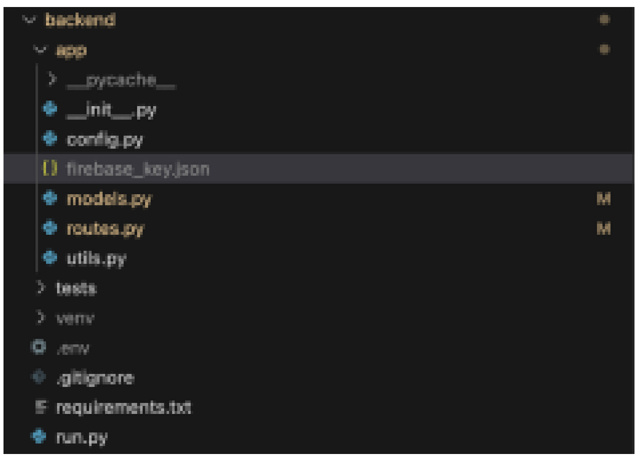
- Select "Start in production mode".
- Click "Create".



3. Set up Service Account Key:
   - Go to Project Settings (⚙️ icon next to "Project Overview").
   - Click Service Accounts.
   - Click "Generate new private key"

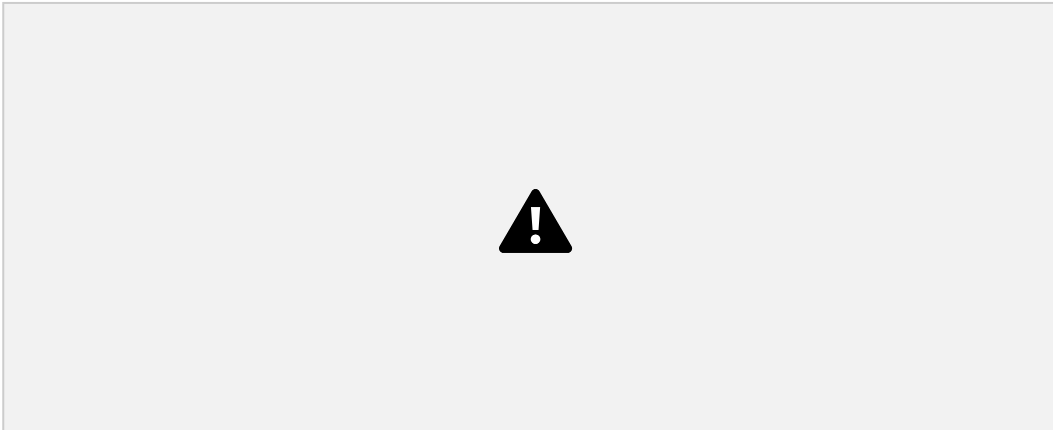


- A JSON file will be downloaded.
- Rename the file to firebase_key.json.
- Move the file into your backend directory: backend/app/firebase_key.json.



Enabling Authentication

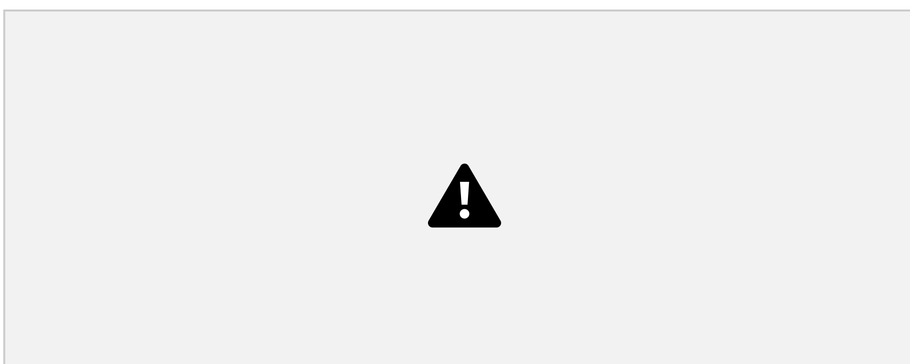
1. In Firebase Console:
   - Navigate to "Build" > "Authentication" > "Sign-in Method".
   - Enable Email/Password sign-in.
   - Click "Save".



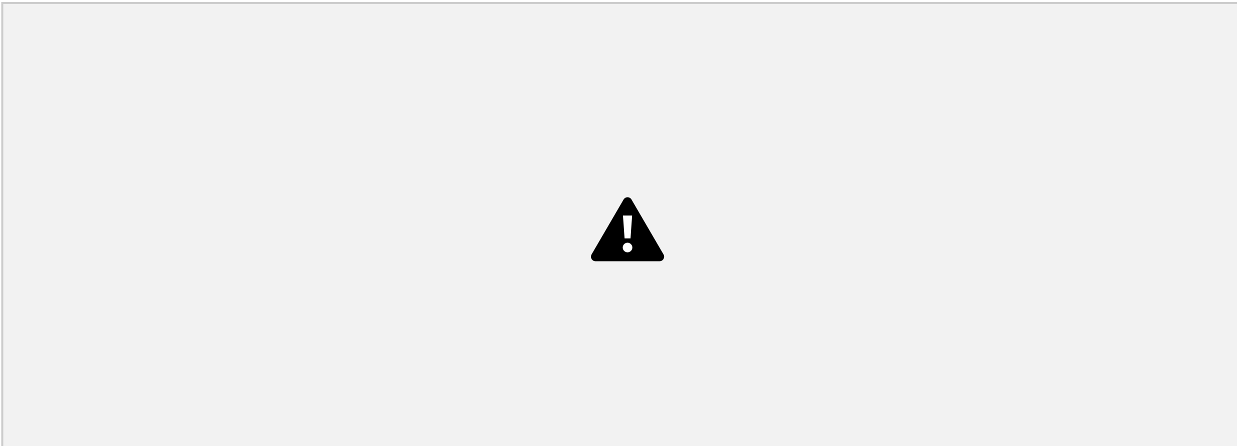
Registering the App on Firebase

1. Go to https://console.firebase.google.com
2. Select your existing project (VAI-DataSystem)
3. In the left sidebar, click Project Settings (⚙️ gear icon)
4. Scroll down to the "Your apps" section
   Click the </> Web icon (for adding a web app)
5. Name your app (VAI-Web), and <u>don't enable Firebase Hosting</u> for now (you can skip that checkbox)
6. Click Register app
7. Copy the `firebaseConfig` and navigate to the frontend folder of your project

   The `firebaseConfig` should look like this:



8. Update your environment file (`.env`) with the details from the `firebaseConfig`



Backend Setup

1. Navigate back to the main project folder:
   cd ..

2. Navigate to the backend folder:
   cd backend

   Create and activate a virtual environment, run this in the terminal:
   python3 -m venv venv

   IMPORTANT NOTE:
   If using Mac or Linux Device:
   - source venv/bin/activate

   If using Windows Device:
   - venv\Scripts\activate

3. Install backend dependencies:
   pip install -r requirements.txt

4. Create a .env file:
   touch .env

   Add the following to your .env file:
   FIREBASE_KEY_PATH=app/firebase_key.json
   FRONTEND_URL=http://localhost:5173

   Ensure your .gitignore includes:
   venv/
   .env
   app/firebase_key.json
   app/__pycache__/

5. Run the backend server:
   python3 run.py

Once you have the frontend, backend, and database locally set up, run the project. Frontend - npm run dev

Backend - python3 run.py

Frontend Deployment (Vite & React)

Deploying to Vercel

1. Go to Vercel.
2. Sign up and connect your GitHub repository.
3. Select the frontend/ folder as the project root.

Add environment variables in the Vercel dashboard under "Settings" > "Environment Variables":
Example:

```
VITE_FIREBASE_API_URL=https://your-production-api-url.com
VITE_FIREBASE_PROJECT_ID=your-production-project-id
```

Vercel will automatically build and deploy the site upon pushing to GitHub.

Local build (Optional):

```
cd frontend
npm run build
```

This generates a dist/ folder with your production static files.

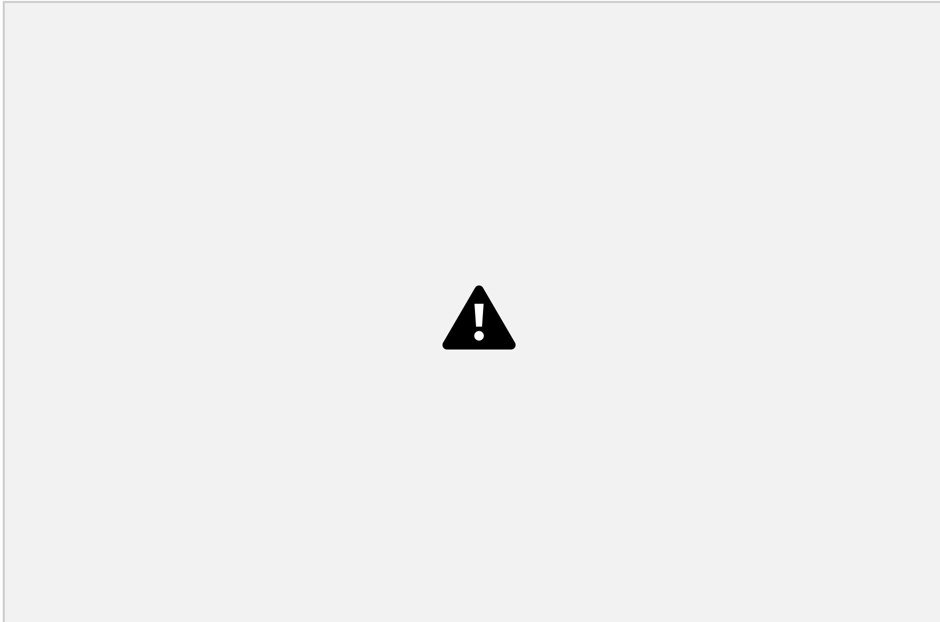
Frontend Deployment (Flask & Python)

Deploying to Render:

1. Go to Render.
2. Sign up and connect your GitHub repository.
3. Create a New Web Service:
   - Root Directory: backend/
   - Environment: Python 3
   - Build Command:
     pip install -r requirements.txt
   - Start Command:
     gunicorn run:app
4. (Install gunicorn by adding it to your requirements.txt if needed.)
5. Add Environment Variables:
   - FIREBASE_KEY_PATH=app/firebase_key.json
   - FRONTEND_URL=https://your-frontend-production-url.com
6. Upload your firebase_key.json securely or manage credentials through Render secrets.

Firebase Setup (Production)

- Ensure Firestore Database Rules are secured.
- Enable Email/Password Authentication.
- Update Authorized Domains under Authentication Settings to include your frontend's production URL.



Additional Notes

• CORS: Install and configure flask-cors if your frontend and backend are on different domains. • Security: Do not expose sensitive keys. Use environment variables.
• Custom Domain (Optional): Configure a custom domain through Vercel and Render settings.

After Deployment

• Verify user login and database reads/writes.
• Confirm frontend and backend URLs are working correctly.
• Monitor deployment logs for any issues.

Rollback Strategy

<u>Frontend (Vercel)</u>

1. Revert to the previous commit or branch in GitHub.
2. Vercel will auto-deploy the last working state.
3. Alternatively, trigger redeployment from Vercel dashboard using the previous deployment.

<u>Backend (Render)</u>

1. In Render dashboard, roll back to the last successful deploy from the "Deploys" tab.
2. Revert GitHub repository to a stable commit and redeploy.
3. Restore previous environment variables if any changes were made.

<u>Firebase</u>

• Revert Firestore rules to a prior saved state if needed.
• Disable or undo changes to authentication methods or authorized domains if problems occur.

PROPOSED WIREFRAMES FOR WEB PAGES VAI STAFF VIEW (FUTURE ENHANCEMENT)

PUBLIC ART INSTALLATION/ NAME

Includes three options:

- Login
- Sign Up
- START

CREATE ACCOUNT/SIGN UP SCREEN

Designated VAI Member will be able to create an account.

LOGIN SCREEN

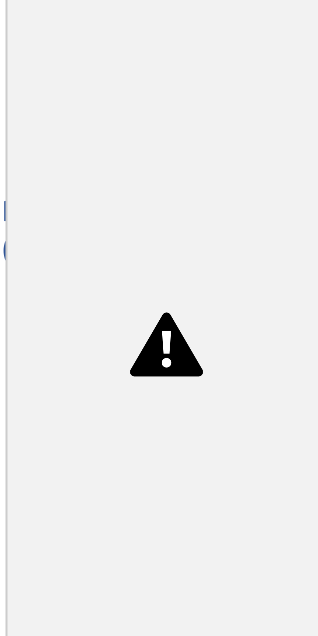
Once an authorized VAI Member has an existing account, the staff member will login with email and password credentials

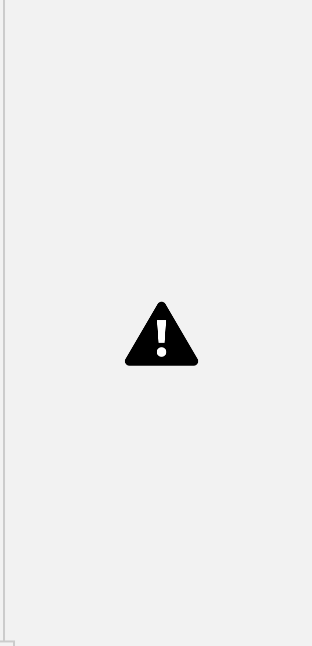
DASHBOARD

Displays live survey results with different options:

- Pie
- Donut
- Bar

** Dashboard offers a "Download" option to save reports and graphs.





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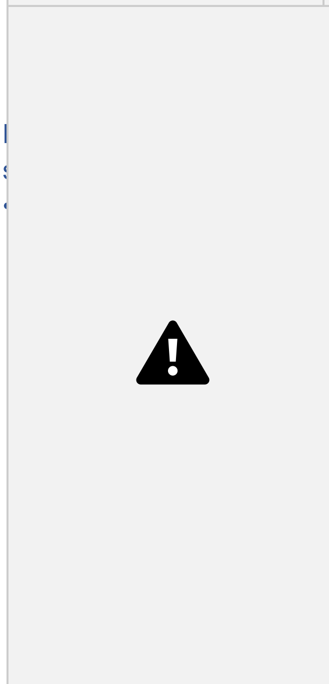

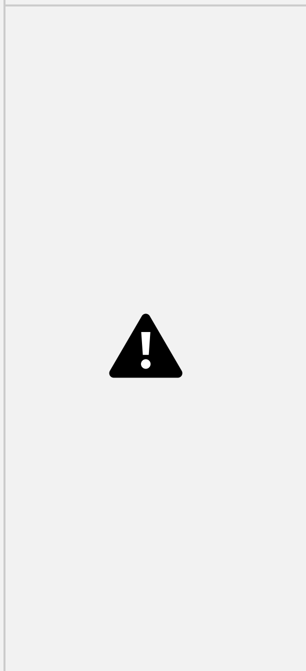

Complete Screen

The participant will be directed to a Thank You page to exit and return to the Home Page

Survey Screen
A survey is recorded once the individual presses Submit at the last question.

The design on the left offers a symmetrical use of whitespace.

Survey Screen
A survey is recorded once the individual presses Submit at the last question.

The design on the right offers a simpler and familiar design

- The survey will be stored once the last question is submitted. A participant would be directed back to the VAI home page once the user presses Done as shown on “Complete Screen” Diagram.

• Diagrams were an early concept of the project but could be implemented in the future

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RESEARCH CITATIONS

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